DexFruit: Dexterous Manipulation and Gaussian Splatting Inspection of Fruit

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***Abstract*—DexFruit is a robotic manipulation framework that enables gentle, autonomous handling of fragile fruit and precise evaluation of damage. Many fruits are fragile and prone to bruising, thus requiring humans to manually harvest them with care. In this work, we demonstrate by using optical tactile sensing, autonomous manipulation of fruit with minimal damage can be achieved. We show that our tactile informed diffusion policies outperform baselines in both reduced bruising and pick-and-place success rate across three fruits: strawberries, tomatoes, and blackberries. In addition, we introduce FruitSplat, a novel technique to represent and quantify visual damage in high-resolution 3D representation via 3D Gaussian Splatting (3DGS). Existing metrics for measuring damage lack quantitative rigor or require expensive equipment. With FruitSplat, we distill a 2D strawberry mask as well as a 2D bruise segmentation mask into the 3DGS representation. Furthermore, this representation is modular and general, compatible with any relevant 2D model. Overall, we demonstrate a 92% grasping policy success rate, up to a 20% reduction in visual bruising, and up to an 31% improvement in grasp success rate on challenging fruit compared to our baselines across our three tested fruits. We rigorously evaluate this result with over 630 trials.**

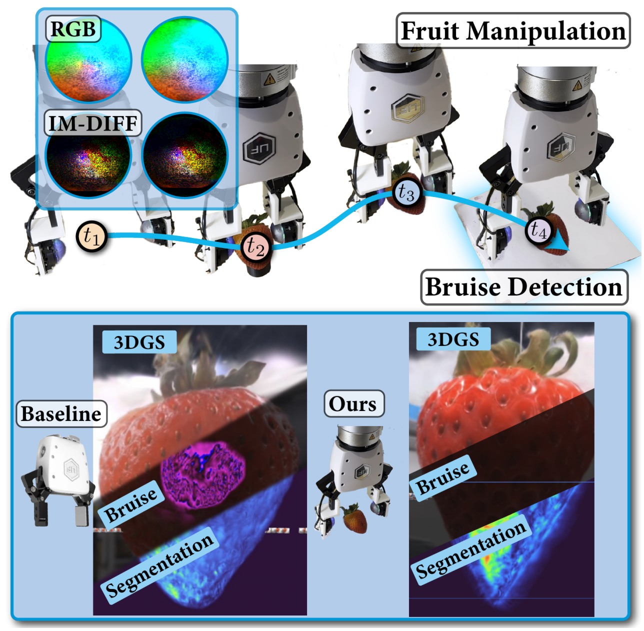

Fig. 1: Our framework, DexFruit, enables safe manipulation of fragile fruit using optical tactile feedback. We further present FruitSplat, a method for accurate 3D reconstruction, automated segmentation, and bruise localization in strawberries.

I. Introduction

Demographic-driven labor shortages and a growing population present significant challenges for the global food supply [1–3]. While some agricultural production is highly mechanized, such as grain and cereal, fruits and vegetables are comparatively limited, often requiring workers to manually harvest produce [1, 4, 5]. Fruit production, in particular, requires a high level of labor-intensive operations to be performed, primarily due to the delicate handling required to preserve quality [4]. Increasing fruit consumption in the United States, coupled with labor-shortage concerns in agriculture, presents a clear need for an automated solution for harvesting delicate fruits [6, 7].

Fruits, such as strawberries, demand significant human labor because of their fragile structure and short post-harvest shelf life; they are typically hand-picked to minimize bruising and decay [4]. Although there are mechanized solutions for fruit and berry harvesting, they often do not meet the quality standards of the fresh market, damage fruit more than their human counterparts [8], and still involve significant human labor in sorting. Robotic solutions for agriculture have long been pursued [9]. Berries have been especially challenging to develop robotic solutions for, as they require extra care and precision in harvesting and manipulation not required for other produce [10–12]. These requirements have still not been fully addressed in the current literature.

Recent advances in gel-based optical tactile sensors, such as DenseTact, have greatly increased the availability of touch for robotics [13–15]. These tactile sensors have been used for many robotic applications, such as manipulation [16, 17], detailed object reconstruction [18, 19] and precise force estimation [20]. Vision-based tactile sensors provide fine-grained tactile information by utilizing a camera sensor that observes the deformation of a transparent gel. In this paper, we apply these tactile sensors to manipulate strawberries, cherry tomatoes, and blackberries with minimal bruising or damage. Strawberries are fragile, vary greatly in size, and show clear bruising on their surface after damage, making them an ideal

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Website: https://dex-fruit.github.io/

challenge fruit for evaluating our method. Cherry tomatoes were selected due to their firm yet delicate skin, which makes them particularly sensitive to improper handling. Blackberries present additional complexity with their soft, easily ruptured structure, necessitating precise manipulation to avoid damage. Together, these fruits provide diverse tactile challenges critical for demonstrating the versatility of our approach. We see this as an important step towards minimizing damage and increasing mechanization in fruit and berry markets.

Adopting robotic solutions for manipulating soft fruits is challenging, primarily due to the difficulty of preventing and accurately measuring bruising and damage caused by robotic grasping [21]. This is complicated by the fact that bruises can be difficult to visually detect or only become noticeable after some time has passed. 3D Gaussian Splatting (3DGS) [22] is a recently developed explicit 3D representation technique, which is capable of training quickly and providing fast and high resolution scene renders. 3DGS has found many uses in robotics, particularly in navigation [23, 24], manipulation [25, 26] and even agriculture [27]. In this paper, we introduce FruitSplat, a new method to accurately visualize and quantify damage in fruit using 3DGS. Existing fruit damage metrics require specialized equipment or rely on overly qualitative human labeling. Motivated by these challenges, our work proposes a method that fuses the 3D accuracy inherent in scanning-based approaches with accessibility of deep learning based approaches. With only a simple web-cam, our approach FruitSplat offers a quantitative, 3D solution for bruise detection, thereby bridging the gap between high-fidelity scanning techniques and traditional deep learning methods.

A. Key Contributions

- We demonstrate the integration of optical tactile sensing into diffusion-based imitation learning policies for soft body manipulation. To the best of our knowledge, our policy is the first to integrate optical tactile sensing with diffusion for fragile fruit manipulation.
- We demonstrate quantitative and qualitative reductions in bruising and improvements in success rate compared to our baselines while using tactile sensing, highlighting the importance of touch for precise manipulation.
- We introduce FruitSplat, a novel method for qualitative and quantitative damage analysis for strawberries. FruitSplat significantly improves on existing evaluations in ease of use and accuracy.

The remainder of the paper is outlined as follows. First Section II addresses related work, section III covers mathematical preliminaries and Section IV introduces our method. Section V showcases our results, and finally, Section VI discusses these results and Section VII presents limitations and future research avenues.

II. Related Work

A. Robotic Fruit Picking

1) Gripper Design: Many works have addressed the use of robotics for fruit picking including apples, strawberries, blackberries [28–33]. These works typically involve the construction of a novel soft gripper, which is unique to a specific fruit or berry. In [28] they focus on apples, and [30, 32] on blackberries. While these grippers generally perform well, they are limited to a single type of fruit. They typically prevent damage through modality specific designs [31, 32] and low dimensional sensing [33]. In this work, we focus on a general gripper capable of manipulating arbitrary fruit. Rather than relying solely on soft robotics techniques or complex designs, we utilize the expressive tactile sensing of DenseTact to prevent damage.

2) Fruit Localization: Fruit localization is a well-studied problem in agricultural robotics. Classical computer vision methods, like RGB based thresholding are applied in [28] to segment apples. More common are deep learning based methods, which use segmentation approaches [30]. More advanced techniques also include 6-DOF 3D pose of fruit [34]. Moreover, new methods have combined these 2D segmentation networks into 3D neural representations [35]. We apply a similar distillation technique using 3DGS for our FruitSplat, while taking advantage of the benefits of Gaussian Splatting.

Overall, fruit localization is a well-solved problem. Instead, we focus our efforts in this paper on manipulation of soft fruits. Rather than focus on fruit picking, we focus on repeated manipulation with minimal damage, which we feel is a more general and a less addressed issue.

B. Bruise Detection and Metrics

Current approaches to bruise detection largely rely on qualitative assessments, with little emphasis on quantitative metrics. For example, [33] employs a visual classification system for raspberries, classifying them as undamaged, mildly damaged, or damaged, which suffers from subjectivity and limited resolution in damage assessment. As quality control of fruit on the market is commonly conducted through visual inspection, allowable fruit quality varies based on inspector [36]. This system is inherently subjective and is in need of a standardized method to quantify damage.

There are a number of quantitative methods for damage detection for agricultural products. [21] provides a comprehensive survey of high-precision methods currently in use. Techniques such as spectroscopy, optical coherence tomography (OCT), magnetic resonance imaging (MRI), and x-ray imaging offer detailed insights into 3D tissue damage, yet their reliance on expensive and complex setups restricts their practical use in robotics. While these techniques offer high resolution and rich information about the quality of a fruit, this is often unnecessary to determine if a fruit should be brought to market.

Deep learning has also been explored to overcome some of these limitations. For example, [37] demonstrated bruise detection on red apples using convolutional models, while [38] addressed strawberry bruise detection aided by UV light. Although these methods enhance detection capabilities compared to purely visual grading, they do not fully leverage the three-dimensional metric information of the fruit.

C. Imitation Learning for Agriculture

Imitation learning is a technique in which robot policies are trained by learning from expert demonstrations. One form of imitation learning which has recently shown strong performance is the diffusion policy [39]. This approach leverages Denoising Diffusion Probabilistic Models (DDPM) [40] to learn observation conditioned, high-dimensional policies. In [41] the authors survey the landscape of imitation learning for agriculture. However they do not find any examples of diffusion policies being used for automated agricultural product grasping. [42] demonstrates force-conditioned tactile and visual policies picking up fragile grapes and placing them on a plate. However, DexFruit not only safely grasps fragile fruit, but we also *quantitatively* measure precisely the damage done to the fruit across hundreds of trials providing a rigorous and necessary analysis of the fruit after manipulation.

D. Optical Tactile Sensing for Agriculture

Recent studies have explored the utility of optical tactile sensing in agricultural applications, particularly for fruit manipulation and evaluation. For instance, [43] employs optical tactile sensing to facilitate fruit manipulation; however, their approach is limited to hard fruits such as apples and lemons, which restricts its applicability across a diverse range of produce. Furthermore, they do not quantify the damage over repeated grasping. In [44] the authors utilize a GelSight [14], vision-based tactile sensor to determine the stiffness of cherry tomatoes. While our current work does not incorporate direct stiffness measurements, the underlying principles of this approach could be integrated into our pipeline. Furthermore, we focus on the manipulation task while preventing damage.

E. Tactile sensing for imitation learning

Recently, several studies have explored incorporating tactile sensing into imitation learning policies, utilizing diverse modalities such as audio [45], low-resolution capacitive sensing [46], and optical tactile sensing [47]. Although these studies demonstrate impressive performance, our work is the first to rigorously validate tactile-feedback-based manipulation of delicate, soft fruits over hundreds of experimental trials.

III. Preliminaries

A. 3D Gaussian Splatting

3D Gaussian Splatting (3DGS) is a powerful 3D visual representation for robotics. Given a series of 100-200 images, 3DGS constructs a visually accurate representation that can be rendered at high FPS on a GPU. We define a ray with camera origin $\mathbf{o}$ and orientation $\mathbf{d}$ as follows:

$$\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}, \quad t \geq 0, \tag{1}$$

The color $C(\mathbf{r})$ along the ray is computed by blending the set of $\mathcal{N}$ ordered points intersecting the ray:

$$\hat{C} = \sum_{i \in N} c_i \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j) \tag{2}$$

where $\alpha_i = 1 - \exp(-\sigma_i \delta_i)$, and d is the distance of a particular point along a ray. The loss used to supervise the Gaussian Splat is a weighted combination of the L1 and SSIM between each ground truth (GT) image and volumetric rendered image.

$$\mathcal{L} = (1 - \lambda)\,\mathcal{L}_1 \;+\; \lambda\,\mathcal{L}_{D\text{-}SSIM} \tag{3}$$

Similar to how we can render color from the ray, we augment additional parameters s_i and b_i to each Gaussian primitive, which dictate if it is a strawberry and is in a bruise region on a per-pixel basis. This can be rendered into mask images and supervised in the same way as color in (2) and (3).

B. Diffusion Policies

Given an observation $\mathcal{O}$, which corresponds to external camera images and robot state, and an action A in the space of robot positions, we seek to learn a policy $\pi : \mathcal{O} \rightarrow \mathcal{P}(A)$. Instead of learning this policy directly we model our policy as a Denoising Diffusion Implicit Model (DDIM) and instead learn a noise predictor network following [39]:

$$\hat{\varepsilon}_k = \varepsilon_\theta(A_t^k, \mathcal{O}_t, k) \tag{4}$$

Where A_t^k is a sequence of noisy actions, $\mathcal{O}_t$ is the observation and denoising iteration k. Given a dataset $\mathcal{D} = \{\{O_t, A_t\}\}_{t=1}^n$ we train this denoising function by applying noise ϵ^k to data A_t and minimizing the following loss:

$$\mathcal{L} = MSE\Big(\epsilon^k, \varepsilon_\theta(A_t^k, \mathcal{O}_t, k)\Big) \tag{5}$$

During training we sample random actions A_t^K and perform denoising using a DDIM scheduler [48].

IV. Method

DexFruit combines a robust manipulation policy and advanced 3D vision techniques to address the challenges of handling delicate fruits in agricultural robotics. We utilize Diffusion Policy trained on expert demonstrations with optical tactile sensing. To quantify fruit damage, we develop FruitSplat, a novel 3D Gaussian Splatting-based analysis tool that accurately reconstructs and compares fruit models before and after manipulation to quantify damage caused during manipulation.

A. Fruit Manipulation Policy

Safely handling fruit in agricultural robotic manipulation requires policies that balance two critical aspects: robustness, to reliably pick and deposit fruits, and sensitivity, to prevent damage during handling. These factors inherently conflict, as robustness encourages a firm grip, whereas damage prevention demands minimal contact. To meet these conflicting demands, we leverage a Diffusion Policy (see Section III), known for smooth and safe execution in complex real-world scenarios. While many existing approaches achieve gentleness through soft gripper hardware, our standard parallel-jaw gripper equipped with DenseTact implicitly achieves softness via the learned policy, successfully minimizing fruit damage.

For this task, our action space consists of the end-effector pose (3D position + 6D rotation [49]) and the width of the gripper normalized between [0,1]. The observation space consists of one RGB camera mounted on the robot arm wrist and RGB images from two DenseTacts. We preform data collection and inference at approximately 10hz. Our RGB and touch images are fed into the policy at a resolution of (224x224). To accentuate the changes in the gripper state during contact, the average channel image difference is computed for each sensor. In addition, we include the current robot state as an observation in the same format as the action. The RGB images are downsampled and encoded into latent vectors using ResNet-34 and concatenated with the state, while the DT images are processed with ResNet-18. Both ResNets are not pretrained. We train the model for 1000 epochs using a DDIM noise scheduler [48]. Expert demonstrations are collected via teleoperation, varying the robot's starting position and individual fruit to ensure the policy conditions on the fruit itself. To ensure consistency across methods and enable a fair comparison with a purely tactile sensing baseline, we fix the initial location of the strawberry. Since diffusion policies have already shown strong capabilities in identifying and grasping objects from diverse positions, we focus exclusively on gentle grasping rather than localization in this work.

During testing, we observed that raw DenseTact images alone provided insufficient signal strength for the policy to effectively leverage tactile information. This was due to minimal variation in the raw images upon contact, making learning challenging. Policies trained without applying image differencing to the DenseTact inputs performed poorly and were therefore not further pursued. Additionally, we noticed that straightforwardly combining vision and tactile images led to worse performance than using either modality individually, likely caused by the model overfitting to correlated image pairs. Inspired by human tactile exploration, we resolved this issue by adaptively switching off visual input when tactile contact occurs, which proved essential for high-performing policies. Contact detection is implemented via a simple energy threshold.

B. FruitSplat

To quantify damage made to strawberries, we leverage 3D Gaussian Splatting to accurately construct strawberries models before and after manipulation. FruitSplat has the following pipeline:

1) Data Pre-Processing: Videos of the entire target strawberry are first collected using a web-cam and then fed directly through COLMAP [50] to obtain sparse points and camera poses of each frame. We then pass each frame into our custom trained YOLOv12 [51] bruising segmentation model for predicted bruise masks. This model was fine-tuned on two thousand hand-labeled images which we collected separately from our experiments. We also apply Grounded SAM 2 (GSAM2) [52] for segmented strawberry masks.

2) 3D Gaussian Splatting: Our custom method directly inherits from Nerfstudio's Splatfacto [53], with "bruise" and"strawberry" added to the list of Gaussian parameters that are learned. Our dataloader utilizes the pre-processed YOLO and GSAM2 masks as pseudo-ground truth for the supervised strawberry and bruise mask learning. Bruise and strawberry loss terms are calculated using cross-entropy loss and are simply added to the main loss. In order to improve training stability we use a stop-grad in the rasterization of bruise and strawberry masks, which only allows gradients with respect to the s_i and b_i parameters of each Gaussian.

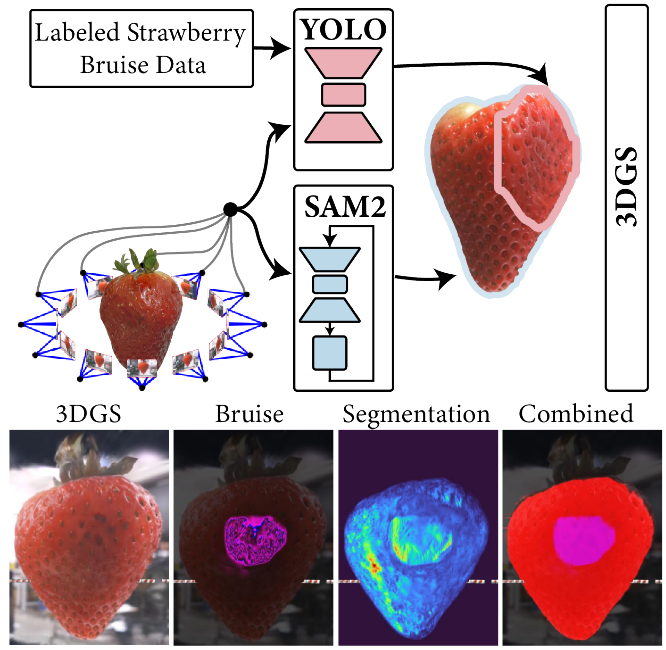

Fig. 2: Here we show the FruitSplat pipeline. Prior to training, the 3D Gaussian Splat the images are processed through 2 models. YOLO segments the bruising, while SAM2 segments the strawberry. These two additional masks are used to supervise additional Gaussian parameters.

3) Post-Processing and Damage Analysis: Once the Gaussian Splat is fully trained after N steps, FruitSplat exports the gaussian means along with the learned gaussian strawberry and gaussian bruise parameters. It then self-filters the point cloud based on the strawberry parameter, leaving a high-resolution point cloud model of the strawberry. Each point is also exported with an encoded bruise score learned from the previously trained 3DGS, which represents the probability of being a bruise. FruitSplat then utilizes the per-point encoded bruise score to calculate the percentage of bruised points that pass a preset confidence threshold, and subsequently compares the difference in bruising between pre-manipulation and post-manipulation strawberry. Due to the fact that the point cloud points already represent only the surface of the strawberry, this method can effectively represent visual surface bruising information. It should be noted that damage analysis requires a pre-manipulation and post-manipulation Gaussian Splat of the strawberry as an input, thus steps 1 and 2 for FruitSplat must be run independently twice. However, both steps only require the use of a camera, with no extra hardware necessary.

V. RESULTS

We demonstrate significant improvements over our baselines in grasping success rate and damage reduction over 630 trials across strawberries, tomatoes and blackberries.

A. Experimental Setup

The experimental setup is shown in Fig. 3. From a starting position, our policy is able to move to the strawberry and a grasp is attempted. The strawberry is then lifted and placed on the green square (goal). Any strawberry that is moved by the policy to the goal is labeled a success, otherwise it is labeled a failure. Failure modalities include missing a grasp of the strawberry completely or dropping the strawberry before reaching the target zone. Following best practices, the strawberries which are used for expert demonstrations are discarded and new strawberries are used for testing. We used the same set of two DenseTact sensors for the entire course of the project, across thousands of demos and experimental roll-outs.

To clearly highlight differences between the evaluated methods, we conduct a series of pick-and-place trials on strawberries, tomatoes, and blackberries. Specifically, we perform 25 trials per strawberry, 50 per tomato, and 20 per blackberry. By conducting such a large number of trials, we both demonstrate the consistency of our method and create a key point of distinction from prior work, while allowing subtle damage to build up and become apparent. For each fruit type, we further vary the size categories: strawberries are tested across three size ranges (small: <4 cm, medium: 4–6 cm, large: >6 cm), tomatoes across two sizes (small: <3 cm, medium: ≥ 3 cm), and blackberries across three sizes (small: <2.5 cm, medium: 2.5–3.5 cm, large: >3.5 cm). These size categories are based on measured fruit length. To ensure consistency, all fruit used in the experiment were sourced on the same day from a single lot, ensuring similar ripeness and quality. The fruit starts on a small 3D-printed pedestal positioned in the center of the white start square. The gripper starts 20 cm away from the strawberry and moves on average 20 cm to the goal region.

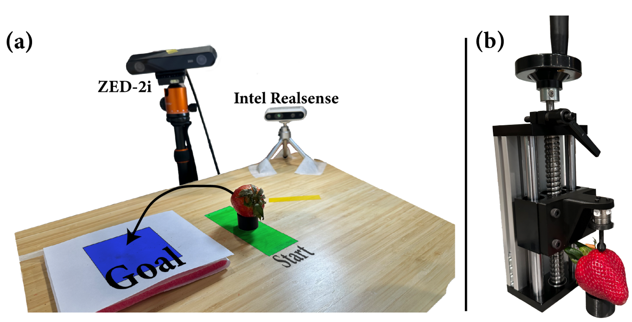

Fig. 3: On the left (a) is the experimental setup. The fruit is initially place in the starting zone on the black pedestal and then lifted to the green goal region. The two sensing modalities used are highlighted in red: the DenseTact optical tactile sensor and IntelRealsense D435i. On the right (b) is the force measuring apparatus which is used to quantify internal damage in the fruit.

We employ an Intel Realsense D435i camera positioned on the robot wrist, angled toward the gripper. We utilize the Ufactory xArm7 operating in position control mode for the experiments.

Data collection is performed via a 3D Connexion space mouse and a mouse-based slider to control gripper position. Additionally, for our full method with tactile sensing, the RGB images of the DenseTact sensors are displayed to the teleoperator. We collect between 100 and 200 demos for each fruit type, ensuring a wide range of variability in our dataset. As our baselines are ablations of the full method we use the same set of demos for each. We take 360-degree videos of each target fruit before and after the manipulations trials. These videos are then fed into our FruitSplat pipeline to produce quantitative damage results.

B. Baselines

In order to effectively evaluate our method, we compare our results to a number of baselines and ablations. *Vision* is our full gripper setup with the DenseTact sensors except the model does not incorporate touch information and instead relies only on the wrist-mounted camera. Likewise *Touch* relies only on the tactile input. Finally we also test a naive combination of vision and touch where both inputs are combined without any natural switching labeled *Vision+Touch**.

C. Metrics

For strawberries, we report a bruising metric to evaluate post-manipulation damage. FruitSplat (as discussed in Section IV) provides a pixel-wise bruising comparison from filtered point-clouds of the trained 3D Gaussian Splats of each strawberry used during experiment trials.

During testing, we found that there were often instances where the internal structure of the fruit was significantly compromised but did not show observable visual changes. In order to measure this internal damage, we assess fruit damage before and after experimentation by measuring the strawberry's ability to resist an external force. We developed a new metric that measures the spring constant of the fruit surface under small deformations. In practice, we find that this accurately measures internal damage to the fruit. We calculate this metric by deforming the fruit a known distance and measuring the associated force required for the deformation. This setup can be seen in Fig. 3. We select equal spaced points along the circumference of each fruit and position the fruit so that the point is normal to a 3D-printed circular indenter. We measure the force in the z-direction after a 1.25mm displacement of z-axis linear guide. The resulting force is measured using an ATI F/T Sensor Nano17. By averaging the spring constant of the four points on each fruit, we compare the effectiveness of the gripper with the DenseTact and its baselines before and after gripping. This metric, *% Stiffness*, is the percent of pre interaction stiffness, which is maintained after the experiment. 100% is the best possible value. Blackberries are excluded from this metric due to their high susceptibility to damage.

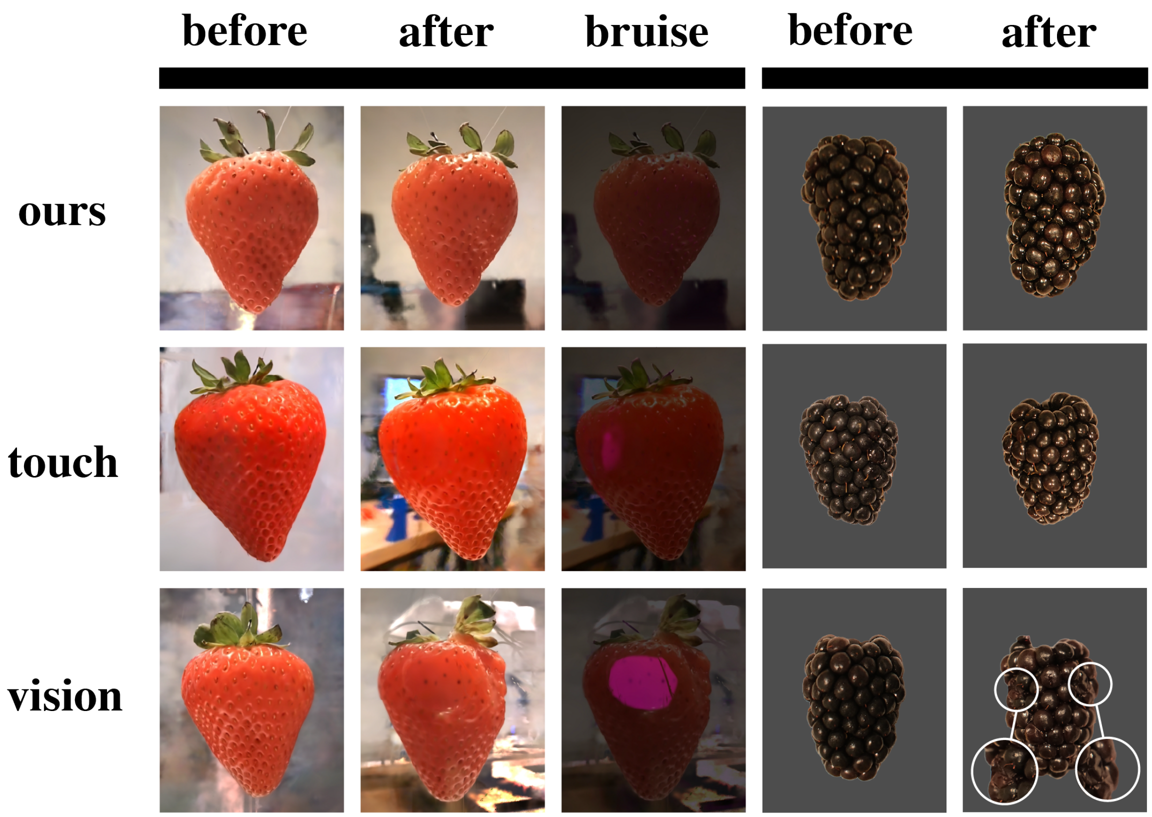

Fig. 4: We show the qualitative results of our method compared to the baselines. On the left, we show the same strawberries before and after the experiments along with the corresponding bruise mask. All the strawberry images are rendered from FruitSplat. Blackberries are shown on the right while tomatoes are not shown as they do not show visible damage.

Method	Strawberry			Tomato		Blackberry
	Success↑	Stiffness↑	Bruise ↓	Success↑	Stiffness↑	Success↑
Ours	**95%**	**80%**	**0.003%**	**100%**	**95%**	**80%**
Touch	67%	78%	**-2.02%**	**100%**	93%	55%
Vision	91%	29%	20.68%	69%	77%	58%

TABLE I: Experimental results. Bold indicates best with margin.

D. *Experimental Results*

We perform a rigorous experimental testing program to effectively compare our method to the baselines of vision-only and touch-only. The quantitative results are shown in Tab. I and Fig. 5, while the qualitative images are shown in Fig. 4. To emphasize the differences between methods we run the same fruit with multiple grasps, 25 for strawberries, 50 for cherry tomatoes and 20 for blackberries. Examining the results in Tab. I, we see that our method outperforms or matches the baselines in success rate across all three of the tested fruits, while also achieving the least damage across our measured metrics. Success is defined as moving the fruit from the start to the goal and is independent from damage. Specifically for strawberries, we found that post-manipulation bruise difference was almost zero, compared to the worst performing baseline of vision only with 20% bruising. For this work, we only implement FruitSplat bruise detection on strawberries, due to its clearly visible external bruises. It should also be noted that the negative percentage in strawberry bruising for touch only is likely caused by slight inconsistencies in pseudo-ground truth bruise mask detection by our YOLO model. We refer the readers to Fig. 5 which includes our estimated error bars based on the variance across trials.

Fig. 5 a provides more detail showing the success rate across the different fruit sizes. We can see that the vision only methods struggle the most on small and medium fruit. We hypothesize that the vision only failure is due to a overfitting on size where the model fits to the average fruit size. While this leads to good grasping success rate on large fruit, it causes grasps which are far to loose on small and medium fruit. Furthermore, this often leads to an overgrasping of large fruit causing more damage. This increased damage is clearly visible on the bottom of Fig. 5. Here we show the % change in stiffness relative to before the experiment. For comparison, we also include both *human* where a human performs the

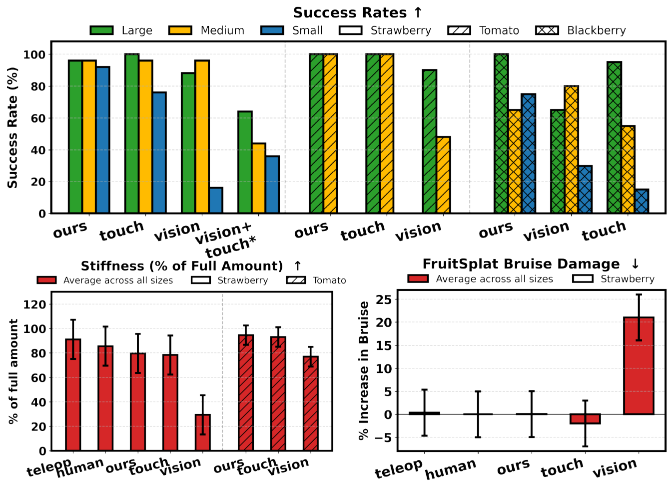

Fig. 5: Above we show the success rates across methods and fruit size. Below on the left we quantify the internal fruit damage measured as a percent change in stiffness, while on the right we show the results from FruitSplat representing external damage.

same action as the robot for each trial, and *teleop* where a human teleoperator controls the robot. While *ours* and *touch* perform similar to the theoretical optimal of *teleop* and *human* the *vision* policy causes a much greater amount of damage. Furthermore, we see that the vision only policy leads to much more external bruising

We can also see in Fig. 5 an additional ablation labeled *vision+touch** which combines vision and touch without any intelligent switching. As discussed in Section V-B this policy performs poorly. We speculate this is due to the policy ignoring the latent tactile vector and overfitting on visual information even after contact has occurred. We can see that unlike the vision or touch only policies, this policy is worse across all three sizes of fruit indicating that it suffers from a different mode of failure.

Finally Fig. 4 shows the qualitative results of our experiments. We can see clear differences in bruising across our method and baselines. The high quality visual accuracy of of the FruitSplat reconstructions of the strawberries are also visible. Tomatoes are not shown as they exhibit nearly no visible bruising when damaged.

VI. Conclusion

This work presents DexFruit, an optical tactile sensing embedded diffusion policy for soft fruit manipulation. We demonstrate that an intelligent switching between visual and tactile inputs greatly improves soft handling of fragile fruits such as strawberries, tomatoes, and blackberries compared to utilizing pure vision or touch sensing. We also present FruitSplat, a novel technique that harnesses the geometric accuracy and photorealism of 3D Gaussian Splatting to qualitatively and quantitatively represent fruit damage post-robot manipulation. Experimentation across various policy ablations and fruit types demonstrate DexFruit's best performance in both policy grasping success rate and post-manipulation damage minimization, demonstrating an average 92% grasping policy success rate, up to a 20% reduction in visual bruising.

VII. Limitations and Future Work

While DexFruit presents improvements in dexterous manipulation of soft fruits, some limitations and several opportunities for future work exist. While in this paper we only applied the FruitSplat bruise detection on strawberries as a representative fragile fruit with visible bruising, in the future, our method could be extended generally to arbitrary fruit types. Our experiments were conducted in controlled laboratory settings rather than actual field conditions. As discussed in Section II we feel that prior work has addressed this challenge well. We believe that integrated ripeness sensing could be achieved by leveraging soft optical tactile sensing output. This system could simultaneously improve yield by reducing damage and improving harvest ripeness. On the policy side, the method for switching off vision and turning on tactile in our method is conditioned on a contact heuristic. While this is sufficient for this work, this could be extended to a learned mode switch as the policy chooses actively which input is more useful.

References

[1] L. Calvin, P. Martin, and S. Simnitt, "Adjusting to higher labor costs in selected u.s. fresh fruit and vegetable industries," U.S. Department of Agriculture, Economic Research Service, Tech. Rep. EIB-235, July 2022.

[2] T. Luo and C. L. Escalante, "Us farm workers: What drives their job retention and work time allocation decisions?" *The Economic and Labour Relations Review*, vol. 28, no. 2, pp. 270–293, 2017.

[3] H. C. J. Godfray, J. R. Beddington, I. R. Crute, L. Haddad, D. Lawrence, J. F. Muir, J. Pretty, S. Robinson, S. M. Thomas, and C. Toulmin, "Food security: the challenge of feeding 9 billion people," *science*, vol. 327, no. 5967, pp. 812–818, 2010.

[4] E. T. Annie Klodd and E. Hoover, "Commercial fruit growing guides: Harvesting strawberries," 2021, accessed: 2025-02-11. [Online]. Available: https://extension.umn.edu/strawberry-farming/harvesting-strawberries

[5] C. W. Bac, E. J. Van Henten, J. Hemming, and Y. Edan, "Harvesting robots for high-value crops: State-of-the-art review and challenges ahead," *Journal of field robotics*, vol. 31, no. 6, pp. 888–911, 2014.

[6] H. Stewart, S. Young, D. Dong, and A. Byrne, "Trends in u.s. fruit consumption relative to recommendations in the dietary guidelines for americans," U.S. Department of Agriculture, Economic Research Service, Tech. Rep. ERR-341, 2024. [Online]. Available: https://doi.org/10.32747/2024.8754754.ers

[7] U.S. Department of Agriculture, Economic Research Service, "Farm labor: Data and research," 2025, accessed: 2025-03-01. [Online]. Available: https://www.ers.usda.gov/topics/farm-economy/farm-labor/

[8] H. Zhou, Y. Zhao, P. Jin, Z. Wang, and Y. Ying, "Intelligent robots for fruit harvesting: recent developments and future challenges," *Precision Agriculture*, vol. 24, no. 2, p. 420–455, 2023.

[9] L. Droukas, Z. Doulgeri, N. L. Tsakiridis, D. Triantafyllou, I. Kleitsiotis, I. Mariolis, D. Giakoumis, D. Tzovaras, D. Kateris, and D. Bochtis, "A Survey of Robotic Harvesting Systems and Enabling Technologies," *Journal of Intelligent & Robotic Systems*, vol. 107, no. 2, p. 21, Jan. 2023. [Online]. Available: https://doi.org/10.1007/s10846-022-01793-z

[10] C. Lenz, R. Menon, M. Schreiber, M. P. Jacob, S. Behnke, and M. Bennewitz, "Hortibot: An adaptive multi-arm system for robotic horticulture of sweet peppers," 2024. [Online]. Available: https://arxiv.org/abs/2403.15306

[11] A. L. Gunderman, J. A. Collins, A. L. Myers, R. T. Threlfall, and Y. Chen, "Tendon-driven soft robotic gripper for blackberry harvesting," *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 2652–2659, 2022.

[12] A. Myers, A. Gunderman, R. Threlfall, and Y. Chen, “Determining hand-harvest parameters and postharvest marketability impacts of fresh-market blackberries to develop a soft-robotic gripper for robotic harvesting,” *HortScience*, vol. 57, pp. 592–594, 05 2022.

[13] W. K. Do, B. Jurewicz, and M. Kennedy, “Densetact 2.0: Optical tactile sensor for shape and force reconstruction,” in *2023 IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 12 549–12 555.

[14] E. Donlon, S. Dong, M. Liu, J. Li, E. Adelson, and A. Rodriguez, “Gelslim: A high-resolution, compact, robust, and calibrated tactile-sensing finger,” in *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2018, pp. 1927–1934.

[15] A. S. V. R. M. N. B. K. S. R. M. H. Q. A. S. B. T. N. T. G. K. D. S. J. K. K. J. K. M. N. S. R. C. T. C.-B. E. S. Y. D. J. M. R. C. Mike Lambeta, Tingfan Wu, “Digitizing touch with an artificial multimodal fingertip,” in *arXiv*, 2024.

[16] W. K. Do, B. Aumann, C. Chungyoun, and M. Kennedy, “Inter-finger small object manipulation with densetact optical tactile sensor,” *IEEE Robotics and Automation Letters*, 2023.

[17] M. Yang, C. Lu, A. Church, Y. Lin, C. Ford, H. Li, E. Psomopoulou, D. A. W. Barton, and N. F. Lepora, “Anyrotate: Gravity-invariant in-hand object rotation with sim-to-real touch,” 2024. [Online]. Available: https://arxiv.org/abs/2405.07391

[18] A. Swann, M. Strong, W. K. Do, G. S. Camps, M. Schwager, and M. Kennedy, “Touch-gs: Visual-tactile supervised 3d gaussian splatting,” in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2024, pp. 10 511–10 518.

[19] M. Strong, B. Lei, A. Swann, W. Jiang, K. Daniilidis, and M. K. III, “Next best sense: Guiding vision and touch with fisherrf for 3d gaussian splatting,” *arXiv*, 2024.

[20] W. K. Do, M. Strong, A. Swann, B. Lei, and M. Kennedy III, “Tensortouch: Calibration of tactile sensors for high resolution stress tensor and deformation for dexterous manipulation,” *arXiv preprint arXiv:2506.08291*, 2025.

[21] Y. He, Q. Xiao, X. Bai, L. Zhou, F. Liu, and C. Zhang, “Recent progress of nondestructive techniques for fruits damage inspection: a review,” *Critical Reviews in Food Science and Nutrition*, vol. 62, no. 20, pp. 5476–5494, 2022.

[22] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, “3d gaussian splatting for real-time radiance field rendering,” *ACM Transactions on Graphics (ToG)*, vol. 42, no. 4, pp. 1–14, 2023.

[23] T. Chen, O. Shorinwa, J. Bruno, A. Swann, J. Yu, W. Zeng, K. Nagami, P. Dames, and M. Schwager, “Splat-nav: Safe real-time robot navigation in gaussian splatting maps,” *arXiv preprint arXiv:2403.02751*, 2024.

[24] T. Chen, A. Swann, J. Yu, O. Shorinwa, R. Murai, M. Kennedy III, and M. Schwager, “Safer-splat: A control barrier function for safe navigation with online gaussian splatting maps,” *arXiv preprint arXiv:2409.09868*, 2024.

[25] O. Shorinwa, J. Tucker, A. Smith, A. Swann, T. Chen, R. Firoozi, M. D. Kennedy, and M. Schwager, “Splat-mover: Multi-stage, open-vocabulary robotic manipulation via editable gaussian splatting,” in *8th Annual Conference on Robot Learning*, 2024.

[26] Y. Zheng, X. Chen, Y. Zheng, S. Gu, R. Yang, B. Jin, P. Li, C. Zhong, Z. Wang, L. Liu, C. Yang, D. Wang, Z. Chen, X. Long, and M. Wang, “Gaussiangrasper: 3d language gaussian splatting for open-vocabulary robotic grasping,” *IEEE Robotics and Automation Letters*, vol. 9, no. 9, pp. 7827–7834, 2024.

[27] T. Ojo, T. La, A. Morton, and I. Stavness, “Splanting: 3d plant capture with gaussian splatting,” in *SIGGRAPH Asia 2024 Technical Communications*, ser. SA ’24. New York, NY, USA: Association for Computing Machinery, 2024.

[28] J. Lv, Y. Wang, L. Xu, Y. Gu, L. Zou, B. Yang, and Z. Ma, “A method to obtain the near-large fruit from apple image in orchard for single-arm apple harvesting robot,” *Scientia horticulturae*, vol. 257, p. 108758, 2019.

[29] Y. Yu, K. Zhang, L. Yang, and D. Zhang, “Fruit detection for strawberry harvesting robot in non-structural environment based on mask-rcnn,” *Computers and electronics in agriculture*, vol. 163, p. 104846, 2019.

[30] A. Qiu, C. Young, A. L. Gunderman, M. Azizkhani, Y. Chen, and A.-P. Hu, “Tendon-driven soft robotic gripper with integrated ripeness sensing for blackberry harvesting,” in *2023 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2023, pp. 11 831–11 837.

[31] C. J. Hohimer, H. Wang, S. Bhusal, J. Miller, C. Mo, and M. Karkee, “Design and field evaluation of a robotic apple harvesting system with a 3d-printed soft-robotic end-effector,” *Transactions of the ASABE*, vol. 62, no. 2, pp. 405–414, 2019.

[32] A. De, D. Kumar, I. Kwuan, A. Qiu, and A.-P. Hu, “Compliant robotic gripper with integrated ripeness sensing for blackberry harvesting,” in *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024, pp. 11 056–11 062.

[33] K. Junge, C. Pires, and J. Hughes, “Lab2field transfer of a robotic raspberry harvester enabled by a soft sensorized physical twin,” *Communications Engineering*, vol. 2, no. 1, p. 40, 2023.

[34] L. Li and H. Kasaei, “Single-shot 6dof pose and 3d size estimation for robotic strawberry harvesting,” 2024. [Online]. Available: https://arxiv.org/abs/2410.03031

[35] L. Meyer, A. Gilson, U. Schmid, and M. Stamminger, “Fruitnerf: A unified neural radiance field based fruit counting framework,” *ArXiv*, February 2024. [Online]. Available: https://meyerls.github.io/fruit_nerf

[36] U. L. Opara and P. B. Pathare, “Bruise damage measurement and analysis of fresh horticultural produce—a review,” *Postharvest Biology and Technology*, vol. 91, pp. 9–24, May 2014. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0925521413003608

[37] Z. Ünal, T. Kızıldeniz, M. Özden, H. Aktaş, and Ö. Karagöz, “Detection of bruises on red apples using deep learning models,” *Scientia Horticulturae*, vol. 329, p. 113021, 2024.

[38] X. Zhou, Y. Ampatzidis, W. S. Lee, C. Zhou, S. Agehara, and J. K. Schueller, “Deep learning-based postharvest strawberry bruise detection under uv and incandescent light,” *Computers and Electronics in Agriculture*, vol. 202, p. 107389, 2022.

[39] C. Chi, Z. Xu, S. Feng, E. Cousineau, Y. Du, B. Burchfiel, R. Tedrake, and S. Song, “Diffusion policy: Visuomotor policy learning via action diffusion,” *The International Journal of Robotics Research*, p. 02783649241273668, 2023.

[40] J. Ho, A. Jain, and P. Abbeel, “Denoising diffusion probabilistic models,” *Advances in neural information processing systems*, vol. 33, pp. 6840–6851, 2020.

[41] S. Mahmoudi, A. Davar, P. Sohrabipour, R. B. Bist, Y. Tao, and D. Wang, “Leveraging imitation learning in agricultural robotics: a comprehensive survey and comparative analysis,” *Frontiers in Robotics and AI*, vol. 11, p. 1441312, 2024.

[42] B. Huang, Y. Wang, X. Yang, Y. Luo, and Y. Li, “3d-vitac: Learning fine-grained manipulation with visuo-tactile sensing,” *arXiv preprint arXiv:2410.24091*, 2024.

[43] Y. Han, K. Yu, R. Batra, N. Boyd, C. Mehta, T. Zhao, Y. She, S. Hutchinson, and Y. Zhao, “Learning generalizable vision-tactile robotic grasping strategy for deformable objects via transformer,” *IEEE/ASME Transactions on Mechatronics*, 2024.

[44] L. He, L. Tao, Z. Ma, X. Du, and W. Wan, “Cherry tomato firmness detection and prediction using a vision-based tactile sensor,” *Journal of Food Measurement and Characterization*, vol. 18, no. 2, pp. 1053–1064, 2024.

[45] Z. Liu, C. Chi, E. Cousineau, N. Kuppuswamy, B. Burchfiel, and S. Song, “Maniwav: Learning robot manipulation from in-the-wild audio-visual data,” *arXiv preprint arXiv:2406.19464*, 2024.

[46] B. Huang, Y. Wang, X. Yang, Y. Luo, and Y. Li, “3d vitac:learning fine-grained manipulation with visuo-tactile sensing.”

[47] S. Jiang, S. Zhao, Y. Fan, and P. Yin, “Gelfusion: Enhancing robotic manipulation under visual constraints via visuotactile fusion,” 2025. [Online]. Available: https://arxiv.org/abs/2505.07455

[48] J. Song, C. Meng, and S. Ermon, “Denoising diffusion implicit models,” in *Proceedings of the 9th International Conference on Learning Representations (ICLR)*, 2021. [Online]. Available: https://openreview.net/forum?id=St1giarCHLP

[49] Y. Zhou, C. Barnes, J. Lu, J. Yang, and H. Li, “On the continuity of rotation representations in neural networks,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 5745–5753.

[50] J. L. Schönberger, E. Zheng, M. Pollefeys, and J.-M. Frahm, “Pixelwise view selection for unstructured multi-view stereo,” in *European Conference on Computer Vision (ECCV)*, 2016.

[51] Y. Tian, Q. Ye, and D. Doermann, “Yolov12: Attention-centric real-time object detectors,” 2025. [Online]. Available: https://arxiv.org/abs/2502.12524

[52] T. Ren, S. Liu, A. Zeng, J. Lin, K. Li, H. Cao, J. Chen, X. Huang, Y. Chen, F. Yan, Z. Zeng, H. Zhang, F. Li, J. Yang, H. Li, Q. Jiang, and L. Zhang, “Grounded sam: Assembling open-world models for diverse visual tasks,” 2024.

[53] M. Tancik, E. Weber, E. Ng, R. Li, B. Yi, J. Kerr, T. Wang, A. Kristoffersen, J. Austin, K. Salahi, A. Ahuja, D. McAllister, and A. Kanazawa, “Nerfstudio: A modular framework for neural radiance field development,” in *ACM SIGGRAPH 2023 Conference Proceedings*, 2023. [Online]. Available: https://dl.acm.org/doi/10.1145/3588432.3591516